

AI-Enabled Autonomous Service Migration for Proactive SLA Assurance in Optical Access and Transport Networks

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Abstract We demonstrate a fully autonomous Network-as-a-Service platform where an AI-assisted orchestrator proactively migrates the services between best-effort and deterministic performance across end-to-end F5G-Advanced access and transport segments to ensure SLA compliance for real-time immersive applications. ©2026 The Author(s)

Overview and Introduction

The evolution of Network-as-a-Service (NaaS) platforms is advancing from programmable provisioning toward fully autonomous operation. Our previous work established a standards-based NaaS framework for F5G-Advanced access and transport networks, enabling reactive Quality-on-Demand (QoD) services through end-to-end (E2E) multi-layer (packet/optical) and multi-domain (access and transport) network slicing with model-driven telemetry [1, 2]. However, this approach requires continuous end-user oversight to authorize transitions between best-effort and premium services, limiting scalability and real-time adaptation. The next step is a closed-loop system in which orchestration intelligence autonomously upgrades services to deterministic transport when performance degrades and reverts them to best-effort when conditions improve, preserving SLAs defined by high-level intents.

This demonstration introduces a novel architectural advancement: a fully autonomous, AI-assisted predictive telemetry analytics integrated into the E2E Service Orchestrator. By embedding a lightweight predictive engine directly into the orchestration logic, the NaaS platform evolves from reactive to proactive and self-optimizing. The innovation lies not in the AI/ML model itself, but in enabling intent-driven, autonomous decision-making that closes the control loop without human intervention. We validate this through a cost-optimized assurance use case for immersive Virtual Reality cloud services, which require low-latency connectivity typically guaranteed by the deterministic fine grain Optical Transport Network (fgOTN) [3] and the deterministic Passive Optical Network (PON) but at higher cost. The orchestrator initially places

services on best-effort access and IP transport, continuously analyzes telemetry, and automatically migrates them to deterministic access and fgOTN upon predicting an SLA violation. It also autonomously reverts traffic to IP transport when conditions improve, ensuring both performance and cost efficiency. This demonstration provides a practical pathway toward zero-touch operation in complex multi-layer networks.

Novelty

This work introduces a foundational shift in NaaS architecture by embedding autonomous intelligence directly into the end-to-end Service Orchestrator. This enables optical networks to evolve from best-effort access and transport pipes into service- and application-aware fabrics capable of learning from real-time network performance and proactively assigning services to best-effort or deterministic slices. The novelty of our demonstration is threefold:

1. *AI-assisted Orchestration Layer for Proactive Assurance*: Unlike conventional telemetry systems that generate alerts only after an SLA violation, our AI-assisted orchestration layer enables autonomous lifecycle management while decoupling platform autonomy from any specific AI model. A lightweight predictive analytics microservice consumes raw telemetry and provides forecasting insights to the orchestrator's decision engine, allowing the underlying algorithms to evolve without modifying the control logic. Telemetry is stored using the IETF SIMAP YANG data model (draft [4, 5]), which we extended to support additional telemetry requirements.
2. *Fully Autonomous Closed-Loop and Predictive Control*: Users specify SLA requirements

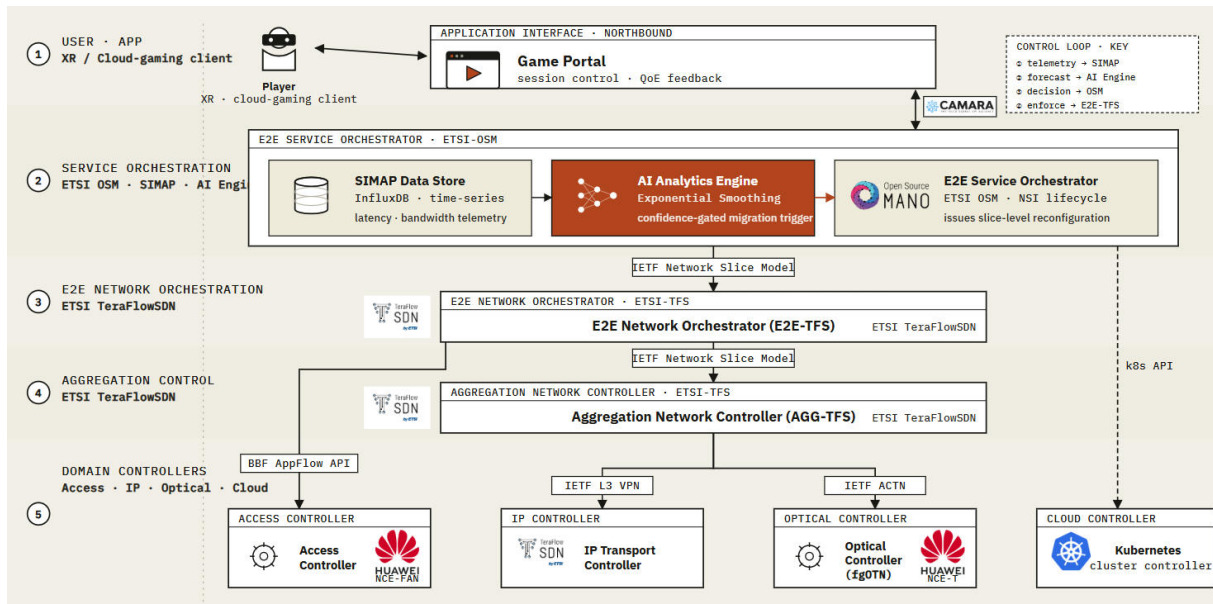


Fig. 1: Hierarchical architecture of the proposed AI-assisted autonomous NaaS framework, spanning the user, service orchestration, network orchestration, aggregation, and domain controller layers.

(e.g., latency $< 15\text{ ms}$), which the system maps into an E2E service configuration. The framework first requests the cloud controller to provision computing resources and instantiate the application server, then instructs the E2E network orchestrator to allocate connectivity over the best-effort PON access and IP transport. Equipped with AI-assisted forecasting, the orchestrator anticipates performance degradation, e.g., predicting congestion minutes in advance and proactively migrating services to deterministic access and fgOTN transport. This shifts operations from “detect-and-fix” to “predict-and-prevent”.

3. *Cost-Aware Dynamic Resource Optimization:* The orchestrator dynamically selects the least expensive slice capable of satisfying the user’s SLA, upgrading to deterministic access and fgOTN transport only when necessary, and reverting to cost-efficient best-effort access and IP transport once performance risk subsides. This aligns operational costs with actual service needs. Fig. 1 illustrates the hierarchical controller architecture, described in detail in the Physical Setup and Equipment section.

ECOC Relevance

This demonstration is highly relevant to the ECOC audience as it presents a tangible implementation of an intelligent control plane that autonomously leverages the unique properties of optical networking, transforming optical access and transport technologies from statically provisioned resources into dynamic, on-demand assets. It provides a concrete example of how predictive analytics can be integrated into network operations to optimize the utilization of high-performance op-

tical resources. The demonstration is built on open-source software (ETSI TeraFlowSDN TFS[6], ETSI OpenSourceMANO OSM[7]) and industry standards (IETF, CAMARA[8], Broadband Forum AppFlow APIs[9]), promoting vendor-agnostic solutions and fostering collaboration. Attendees will see a research-stage prototype of a fully autonomous optical network, stimulating important discussions on architectural requirements, data models, AI-driven operations, and the practical challenges of zero-touch operation in multi-layer optical networks.

Demo Content and Implementation

Objectives and Configuration:

The primary objective is to demonstrate the platform’s fully autonomous closed-loop control in a realistic scenario. We configure a cloud gaming session with a strict latency SLA ($< 15\text{ ms}$). The user provides a high-level intent specifying this performance target. The system’s goal is to maintain this SLA autonomously by managing the E2E connectivity service: it begins on the best-effort access flow and IP transport network and migrates to a deterministic access and fgOTN-based transport network only when necessary to avoid predicted SLA violations.

Physical Setup and Equipment:

The demonstration runs on our established multi-domain testbed, whose hierarchical architecture is shown in Fig. 1 and described in detail in our prior work [1, 2]. The access segment uses Huawei OLT/ONTs with FTTR devices supporting best-effort and deterministic PON slicing, controlled by Huawei’s NCE-FAN controller via Broadband Forum AppFlow APIs [9]. The transport layer comprises two parallel domains: a best-effort IP net-

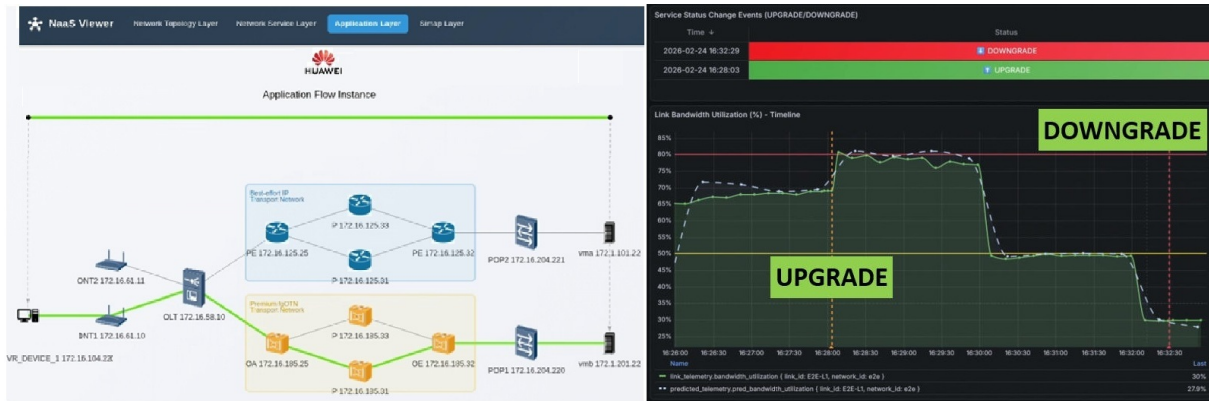


Fig. 2: (Left): Our Naas Viewer. (Right): The Grafana dashboard showing the real-time and predicted bandwidth utilization and AI-based autonomous decisions.

work, provided by Telefonica and managed by a dedicated ETSI TeraFlowSDN instance (IP-TFS), and a deterministic fgOTN network managed by Huawei's NCE-T controller via IETF ACTN [10], both coordinated by an Aggregation Network Controller (AGG-TFS) that exposes the IETF Network Slice model [11] northbound to the E2E Network Orchestrator (E2E-TFS). A Kubernetes cluster hosts the cloud gaming application server. The control plane is built on ETSI OSM as the E2E Service Orchestrator, into which we integrate two new components: a SIMAP Data Store implemented on a time-series database (InfluxDB) that stores per-link latency and bandwidth telemetry tagged with service-specific metadata and hierarchically aggregated into an E2E service view, and an AI Analytics Engine that applies Exponential Smoothing over the historical time-series to produce short-horizon forecasts with associated confidence indicators. The modular design of this AI Analytics Engine permits more advanced ML models to be substituted without altering the orchestration logic. OSM uses these forecasts to drive confidence-gated migration decisions: when the predicted latency is forecast to breach the SLA within a configurable look-ahead window (e.g., 90 s) at sufficient confidence, OSM issues an updated IETF Network Slice request, and E2E-TFS, via AGG-TFS and NCE-T, establishes a deterministic PON and fgOTN lightpath replacing the best-effort PON and IP path for the affected flow; low-confidence forecasts preserve the current slice to avoid unnecessary churn.

Presentation to Attendees and Attendees Interaction:

The demonstration will be conducted live via a VPN connection to the lab testbed. Three integrated views will be presented. First, our graphical topology viewer displays the live network topology and the active end-to-end path of the user's session, visually distinguishing whether traffic is traversing the best-effort PON and IP transport or the deterministic PON and fgOTN transport.

A screenshot of our implemented NaaS viewer is shown in Fig. 2, left. Second, a Grafana-based telemetry dashboard presents real-time time-series graphs of latency and bandwidth utilization alongside the AI-generated prediction overlays, giving attendees direct visibility into the forecasting horizon. An accompanying event log surfaces autonomous orchestration decisions as they occur, e.g., flagging an imminent SLA breach and confirming a subsequent upgrade or downgrade action. A sample of this dashboard is shown in Fig. 2, right. Third, a game portal interface represents the user-facing layer of the NaaS platform, through which a user selects a cloud-based AR/VR gaming application, each carrying its own SLA profile. Once the session is instantiated, the portal reflects the active slice type in real time, updating automatically as the framework autonomously migrates the service between transport tiers. Attendees will observe the full closed-loop sequence: a predicted latency exceedance, a seamless migration to deterministic PON and fgOTN transport, SLA preservation, and an autonomous reversion to best-effort PON and IP transport once conditions stabilize. To make the impact of autonomous service migration tangible, the demonstration also features a custom-built interactive VR tennis game, developed in-house and running live. Attendees are invited to wear a VR headset and play a session while the demo deliberately drives the underlying IP transport toward congestion. Participants will directly experience the perceptual effects of degradation on the best-effort path, i.e., increased motion-to-photon delay and reduced responsiveness, followed by immediate quality recovery as the AI-assisted orchestrator predicts the impending SLA breach and seamlessly migrates the session onto the deterministic fgOTN transport. This hands-on element allows attendees to feel the practical difference between best-effort and deterministic optical transport and to appreciate the end-user value delivered by predictive, intent-driven closed-loop orchestration.

Acknowledgements

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